

The empirical recurrence for OEIS A254588, recovered from its tail

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Abstract

OEIS A254588 records “Empirical recurrence of order 66” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 331$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 16$, so the recurrence is proved for every $n \geq 82$. Its last coefficient is $c_{66} = 137438953472 \neq 0$, so no further tail satisfies anything shorter; any order-66 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A254588 is “Number of $(3+2)X(n+2)$ 0..1 arrays with every $3X3$ subblock sum of the two medians of the central row and the central column and the two maximums of the diagonal and antidiagonal nondecreasing horizontally, vertically and ne-to-sw antidiagonally.”. It has offset 1 and begins

15051, 110259, 774168, 5909527, 45867931, 350562134, 2634910318, 19541565637, ...

The entry states:

Empirical recurrence of order 66 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 14:51:40, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 331$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 331$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 66: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 16$ is the first whose minimal order is exactly the stated 66.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 662$ exact terms from $n = 16$ on, it returns order 66 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{66} c_i a(n-i),$$

c_1	15	c_{18}	−982605097	c_{35}	5235981008048	c_{52}	−78146546434048
c_2	−43	c_{19}	−525661827	c_{36}	−8961320875968	c_{53}	90659961176064
c_3	−301	c_{20}	3537210873	c_{37}	9495076668384	c_{54}	−54038659858432
c_4	1733	c_{21}	−13604118395	c_{38}	−3340248323904	c_{55}	−407025745920
c_5	−1411	c_{22}	26058085893	c_{39}	−7511230200448	c_{56}	38103729307648
c_6	−3935	c_{23}	−21585374861	c_{40}	18343060535296	c_{57}	−43083597086720
c_7	30651	c_{24}	11583339	c_{41}	−21290971470080	c_{58}	23873810595840
c_8	−214543	c_{25}	88700798091	c_{42}	7684278387200	c_{59}	−1679936192512
c_9	352465	c_{26}	−261763128687	c_{43}	19145034636288	c_{60}	−9753468076032
c_{10}	675875	c_{27}	370621113185	c_{44}	−46219530362880	c_{61}	9493756772352
c_{11}	−2626489	c_{28}	−364080189946	c_{45}	53067439591424	c_{62}	−4138200989696
c_{12}	9819780	c_{29}	−14167508972	c_{46}	−23766139846656	c_{63}	62277025792
c_{13}	−26022750	c_{30}	1053920264814	c_{47}	−31571693314048	c_{64}	1090921693184
c_{14}	−807532	c_{31}	−2155578606768	c_{48}	83059134545920	c_{65}	−652835028992
c_{15}	90973530	c_{32}	2844820069284	c_{49}	−95505387749376	c_{66}	137438953472
c_{16}	−278613639	c_{33}	−2245208451152	c_{50}	53673269985280		
c_{17}	862079053	c_{34}	−902781384392	c_{51}	18533572935680		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 82$, and it is the recurrence OEIS A254588 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 16$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{66} - c_1x^{66-1} - \dots - c_{66}$. Its constant term is $-c_{66} = -137438953472$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 66 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 66, so $\deg q = 66$, and it is monic. It holds from some index t on. From $\max(t, 16)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 66, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 16 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 66, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 137438953472.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-66 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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